

Show understanding of the limitations of using a file-based approach for the storage and retrieval of data

9608_s21_qp_12_sc_4.a

QUESTION

- 4 K2 Mountain Guiding is a company that runs courses teaching people how to climb mountains.

The company uses a file-based approach to store and retrieve data.

- (a) Describe **three** drawbacks of a file-based approach compared to a relational database.

Drawback 1

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Drawback 2

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Drawback 3

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[6]

MARK SCHEME

4(a)	1 mark per drawback identified, 1 mark for related description (max 6) • There is data duplication // redundant data • ...because the same data is stored multiple times // data changed in one file is not automatically changed in others // by example • There could be data inconsistency // reduced data integrity • ... because duplicated data might be stored as different values // by example • There is program-data dependency • ... if the data structure changes all the programs accessing that data must be changed too // by example • It is not easy to perform complex searches/queries • ... a new program has to be written each time • Lack of privacy • ...as access controls are usually to the system rather than the data // user views cannot easily be implemented	6
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9608_s20_qp_11_sc_7.a

QUESTION

- 7 A driving school teaches people how to drive cars. The school has a relational database, DRIVING_SCHOOL, to store information about instructors, students, lessons and the cars used by instructors.

INSTRUCTOR(InstructorID, FirstName, LastName, DateOfBirth, Level)

CAR(Registration, Make, Model, EngineSize)

INSTRUCTOR_CAR(InstructorID, Registration)

STUDENT(StudentID, FirstName, LastName, DateOfBirth, Address1)

LESSON(LessonID, StudentID, InstructorID, LessonDate, LessonTime)

- (a) Give **two** benefits to the **driving school** of using a relational database instead of a flat file.

1.....

.....

2.....

.....

[2]

MARK SCHEME

7(a)	1 mark per bullet point to max 2 • Reduced data redundancy • Reduced data dependency • Improved data integrity • Improved data privacy • Program-data independence • Ability to create ad hoc queries	2
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9608_w19_qp_12_sc_4.a.i

QUESTION

4 Anushka needs to store information about bookings at a sports club.

(a) Anushka has a file-based storage system. She wants a relational database.

(i) Describe the features of a relational database that address the limitations of Anushka's file-based system.

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..... [4]

MARK SCHEME

4(a)(i)	1 mark per bullet point, max 3 marks from any group to max 4 Multiple tables are linked together ... which eliminates / reduces data redundancy / duplication ... and increases <u>data</u> integrity / consistency ... which reduces compatibility issues ... so data need only be updated once ... and associated data will be automatically updated // referential integrity can be enforced ... which eliminates unproductive maintenance // which makes it easier to maintain the data Program-data independence means that ... the structure of data can change and does not affect program ... the structure of programs can change and does not affect data ... the data can be accessed by any appropriate program Allows concurrent access to data ... by the use of record locking ... by restricting over-writing changes Complex queries can be more easily written ... to search / find specific data // specific example related to the sports club Different users can be given different access rights ... which improves security Different users can be given different views of the data ... so they do not see confidential information ... and data privacy is maintained ... accept a valid example related to the sports club	4
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9608_s19_qp_12_sc_5.a.i

QUESTION

5 Moheem is creating a relational database to store data about his customers. (a) Moheem has been told a relational database addresses some of the limitations of a file-based approach by reducing data redundancy. (i) State what is meant by the term data redundancy. [1]
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MARK SCHEME

5(a)(i)	1 mark for correct answer <u>Repeated</u> / <u>duplicated</u> data	1
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QUESTION

(b) The software company stores information about customers and the software licences they have purchased. The company considers a file-based approach for the storage and retrieval of data.

(i) Give **three** limitations of a file-based approach to store the data.

- 1
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- 2
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- 3
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[3]

MARK SCHEME

2(b)(i)	1 mark per bullet point to max 3 Data redundancy // data is repeated in more than one file Data dependency // changes to data means changes to programs accessing that data Lack of data integrity // entries that should be the same can be different in different places Lack of data privacy // all users have access to all data if a single flat file	3
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QUESTION

9 A health club offers classes to its members. A member needs to book into each class in advance.

- (a) The health club employs a programmer to update the class booking system. The programmer has to decide how to store the records. The choice is between using a relational database or a file-based approach.

Give **three** reasons why the programmer should use a relational database.

- 1
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.....[6]

MARK SCHEME

9 (a) ONE mark for each reason and **ONE** mark for a further explanation. **MAX THREE** reasons.

- Reduced data redundancy / data duplication
- Data is stored in (separate) linked tables
- The database (generally) stores data only once / data need only be updated once
- Improved data consistency / integrity / associated data will be automatically updated / easier to maintain the data / elimination of unproductive maintenance
- Complex queries can be more easily written
- To search / find specific data // specific example related to the Health Club
- Fields can be more easily added to or removed from tables
- Without affecting existing applications (that do not use these fields)
- Program-data dependence is overcome
- Changes to the data (design) do not require changes to programs // changes to programs do not require changes to data // the data can be accessed by any appropriate program
- Security is improved
- Each application only has access to the fields it needs // different users can be given different access rights
- Different users can be given different views of the data / data privacy is maintained
- So they do not see confidential information
- Allows concurrent access
- Record locking prevents two users updating the same record at the same time // record locking assures data consistency

[6]

9608_s16_qp_12_sc_8.a.iv

QUESTION

(iv) The DBMS has replaced software that used a file-based approach with a relational database.

Describe how using a relational database has overcome the previous problems associated with a file-based approach.

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.....[3]

MARK SCHEME

(iv) Three points from:

[3]

- By storing data in (separate) linked tables data redundancy is reduced / data duplication is controlled...
- Compatibility / data integrity issues are reduced as data only needs to be updated once / is only stored once.
- Unwanted or accidental deletion of linked data is prevented as the DBMS will flag an error.
- Program - data dependence is overcome.
- Changes made to the structure of the data have little effect on existing programs.
- Ad-hoc / complex queries can be more easily made as the DBMS will have a query language/ QBE form.
- Unproductive maintenance is eliminated as changes only need to be made once (rather than changing multiple programs).
- Fields can be added or removed without any effect on existing programs (that do not use these fields).
- Security / privacy of the data is improved as each application only has access to the fields it needs.
- There is better control of data integrity as the DBMS (uses its Data Dictionary) to perform validation checks on data entered.

9608_s16_qp_11_sc_8.a.iv

QUESTION

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- There is better control of data integrity as the DBMS (uses its Data Dictionary) to perform validation checks on data entered.

Describe the features of a relational database that address the limitations of a file-based approach

9608_s20_qp_11_sc_7.a

QUESTION

- 7 A driving school teaches people how to drive cars. The school has a relational database, DRIVING_SCHOOL, to store information about instructors, students, lessons and the cars used by instructors.

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STUDENT(StudentID, FirstName, LastName, DateOfBirth, Address1)

LESSON(LessonID, StudentID, InstructorID, LessonDate, LessonTime)

- (a) Give **two** benefits to the **driving school** of using a relational database instead of a flat file.

- 1.....
.....
2.....
.....

[2]

MARK SCHEME

7(a)	1 mark per bullet point to max 2 • Reduced data redundancy • Reduced data dependency • Improved data integrity • Improved data privacy • Program-data independence • Ability to create ad hoc queries	2
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QUESTION

4 Anushka needs to store information about bookings at a sports club.

(a) Anushka has a file-based storage system. She wants a relational database.

(i) Describe the features of a relational database that address the limitations of Anushka's file-based system.

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9608_s19_qp_12_sc_5.a.ii

QUESTION

(ii) Explain **how** a relational database can help to reduce data redundancy.

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..... [3]

MARK SCHEME

5(a)(ii)	1 mark per bullet point • Because each record/piece of data is stored once <u>and</u> is referenced by a (primary) key • Because data is stored in individual tables • ...and the tables are linked by relationships • By the proper use of Primary and Foreign keys • By enforcing referential integrity • By going through the normalisation process	3
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9608_w16_qp_12_sc_9.a

QUESTION

9 A health club offers classes to its members. A member needs to book into each class in advance.

- (a)** The health club employs a programmer to update the class booking system. The programmer has to decide how to store the records. The choice is between using a relational database or a file-based approach.

Give **three** reasons why the programmer should use a relational database.

1

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2

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[6]

9608_s16_qp_12_sc_8.a.iv

QUESTION

- (iv)** The DBMS has replaced software that used a file-based approach with a relational database.

Describe how using a relational database has overcome the previous problems associated with a file-based approach.

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.....[3]

MARK SCHEME

(iv) Three points from:

[3]

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9608_s16_qp_11_sc_8.a.iv

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Show understanding of and use the terminology associated with a relational database model

9608_s21_qp_13_sc_8.b

QUESTION

(b) Explain how primary and foreign keys are used to link the tables in CHARITY_PROJECT.

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..... [4]

MARK SCHEME

8(b)	1 mark per bullet point • EmployeeID is the Primary Key in EMPLOYEE • ...links to EmployeeID which is the Foreign Key in PROJECTTEAM • ProjectID is the Primary Key in PROJECT • ...links to ProjectID which is the Foreign Key in PROJECTTEAM	4
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QUESTION

- 5 A company runs activity courses. It is creating a relational database to store details of the courses it runs.

The database has five tables:

EMPLOYEE(EmployeeID, FirstName, LastName, Role, Language)

BOOKING(BookingID, CustomerID, CourseID)

CUSTOMER(CustomerID, FirstName, LastName)

COURSE(CourseID, Title, Level, Date)

COURSE_EMPLOYEE(CourseID, EmployeeID)

- (a) Complete the entity relationship (E-R) diagram for the database.



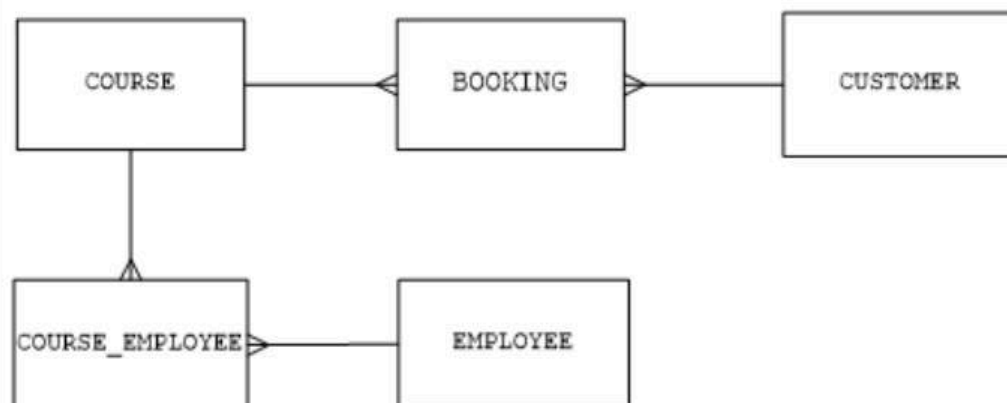
[4]

MARK SCHEME

5(a)

1 mark for each link:

- 1 CUSTOMER to many BOOKING
- 1 COURSE to many BOOKING
- 1 COURSE to many COURSE_EMPLOYEE
- 1 EMPLOYEE to many COURSE_EMPLOYEE



4

QUESTION

- (b) Describe what is meant by **referential integrity**.
Give an example from the `CUSTOMER` and `BOOKING` tables in your answer.

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.....

.....

..... [2]

MARK SCHEME

5(b)	1 mark for description and 1 mark for application to the given tables e.g. (2 marks) - each value stored in the <code>CustomerID</code> (FK) field in the <code>Booking</code> table must have a corresponding value (1) in the <code>CustomerID</code> (PK) field in the <code>Customer</code> table (1) e.g. (1 mark) - Each foreign key value must have a matching value in the primary key of the linked table (1)	2
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QUESTION

6 Sheila creates a relational database for her hotel using a Database Management System (DBMS).

(a) Draw **one** line from each database term to its most appropriate description.

Database Term	Description
Primary key	A field in one table that links to a primary key in another table
Attribute	A collection of records and fields
Foreign key	The type of data that is being stored
Entity	A unique identifier for each tuple
	A data item, represented as a field within a table
	The concept or object in the system that we want to model and store information about

[4]

MARK SCHEME

6(a)	1 mark for each correct line	4
Database Term		Description
Primary key		A field in one table that links to a primary key in another table
Attribute		A collection of records and tables
Foreign key		The type of data that is being stored
Entity		A unique identifier for each tuple
		A data item, represented as a field within a table
		The concept or object in the system that we want to model and store information about

9608_s20_qp_12_sc_6.b

QUESTION

(b) Give an example from the database `SOFTWARE_MANAGEMENT` for each of the following database terms.

Term	Example
Entity	
Foreign key	
Attribute	

[3]

MARK SCHEME

6(b)	1 mark for a suitable example for each	3								
	<table><tr><th>Term</th><th>Example</th></tr><tr><td>Entity</td><td>SOFTWARE_PURCHASED //CUSTOMER_DETAILS</td></tr><tr><td>Foreign Key</td><td>CustomerID (in SOFTWARE_PURCHASED table)</td></tr><tr><td>Attribute</td><td>Any valid example of an attribute from the tables</td></tr></table>	Term	Example	Entity	SOFTWARE_PURCHASED //CUSTOMER_DETAILS	Foreign Key	CustomerID (in SOFTWARE_PURCHASED table)	Attribute	Any valid example of an attribute from the tables	
Term	Example									
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Foreign Key	CustomerID (in SOFTWARE_PURCHASED table)									
Attribute	Any valid example of an attribute from the tables									

9608_w19_qp_13_sc_3.a.i

QUESTION

3 A hotel needs to record information about customers and their bookings.

(a) The hotel has two types of room: double and family. Each room has a unique room number.

The hotel stores information about the customers including their name, address and contact details.

When a customer books a room, they give the start date and the number of nights they want to stay. If a customer wants more than one room, each room must have a separate booking. Each booking has an ID number.

The hotel creates a normalised, relational database to store the required information. There are three tables:

- CUSTOMER
- ROOM
- BOOKING

(i) Complete the database design for the hotel by writing the attributes for each table.

CUSTOMER (.....

)
 ROOM (.....

)
 BOOKING (.....

)

[3]

MARK SCHEME

3(a)(i)	1 mark per table CUSTOMER table has at least customer ID, customer name, address and contact details ROOM has at least room number, room type, BOOKING has at least booking ID, room number, customer ID, start date, number of nights CUSTOMER (<u>CustomerID</u>, Name, Address, ContactDetails) ROOM (RoomNumber, RoomType) BOOKING (BookingID, RoomNumber, CustomerID, StartDate, NumberNights)	3
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9608_w19_qp_13_sc_3.a.ii

QUESTION

(ii) Identify the primary key for each table that you designed in **part (a)(i)**.

CUSTOMER

ROOM

BOOKING [2]

MARK SCHEME

3(a)(ii)	1 mark for 1 or 2 correct Primary Keys, 2 marks for 3 correct Primary Keys CUSTOMER: CustomerID ROOM: RoomNumber BOOKING: BookingID	2
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9608_w19_qp_13_sc_3.a.iii

QUESTION

(iii) Identify **one** foreign key in the tables that you designed in **part (a)(i)**.

Table name

Foreign key [1]

MARK SCHEME

3(a)(iii)	1 mark for both table name and Foreign Key Table: <code>BOOKING</code> Foreign Key: <code>CustomerID / RoomNumber</code>	1
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9608_w19_qp_11_sc_4.d.i

QUESTION

- (d) The bank uses a relational database, `ACCOUNTS`, to store the information about customers and their accounts.

The database stores the customer's first name, last name and date of birth.

The bank has several different types of account. Each account type has a unique ID number, name (for example, regular or saving) and bonus (for example, \$5.00, \$10.00 or \$15.00).

A customer can have more than one account.

Each customer's account has its own ID number and stores the amount of money the customer has in that account.

The bank creates a normalised, relational database to store the required information. There are three tables:

- `CUSTOMER`
- `ACCOUNT_TYPE`
- `CUSTOMER_ACCOUNT`

- (i) Write the attributes for each table to complete the database design for the bank.

```

CUSTOMER ( .....
.....
.....)

ACCOUNT_TYPE ( .....
.....
.....)

CUSTOMER_ACCOUNT ( .....
.....
.....)
[3]
  
```

MARK SCHEME

4(d)(i)	1 mark per table Table CUSTOMER with fields FirstName, LastName, DateOfBirth, CustomerID Table ACCOUNT_TYPE with fields AccountID, Name, Bonus Table CUSTOMER_ACCOUNT with fields ID, CustomerID, AccountID, Amount CUSTOMER (<u>CustomerID</u>, FirstName, LastName, DateOfBirth) ACCOUNT_TYPE (<u>AccountID</u>, Name, Bonus) CUSTOMER_ACCOUNT (<u>ID</u>, CustomerID, AccountID, Amount)	3
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9608_w19_qp_11_sc_4.d.ii

QUESTION

(ii) Identify the primary key for each table that you designed in **part (d)(i)**.

CUSTOMER

ACCOUNT_TYPE

CUSTOMER_ACCOUNT [2]

MARK SCHEME

4(d)(ii)	1 mark for 1 or 2 correct Primary Keys, 2 marks for 3 correct Primary Keys CUSTOMER: CustomerID ACCOUNT_TYPE: AccountID CUSTOMER_ACCOUNT: ID	2
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9608_w19_qp_11_sc_4.d.iii

QUESTION

(iii) Identify **one** foreign key in one of the tables that you designed in **part (d)(i)**.

Table name

Foreign key [1]

MARK SCHEME

4(d)(iii)	1 mark for both table name and Foreign Key Table: CUSTOMER_ACCOUNT Foreign Key: CustomerID / AccountID	1
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9608_w19_qp_11_sc_4.d.iv

QUESTION

(iv) The following table has definitions of database terms.

Write the correct database term in the table for each definition.

Definition	Term
All the data about one entity	
The data in one row of a table	
A column or field in a table	

[3]

MARK SCHEME

4(d)(iv)	1 mark for each correct term		3
	Definition	Term	
	All the data about one entity	Table / Relation	
	The data in one row of a table	Tuple / Record	
	A column or field in a table	Attribute	

9608_s19_qp_13_sc_3.b

QUESTION

(b) Relationships are created between tables using primary and foreign keys.

Describe the role of a primary and a foreign key in database relationships.

.....

 [2]

MARK SCHEME

3(b)	1 mark per bullet point to max 2 Primary key uniquely identifies each tuple // Each tuple in the table is unique Primary key can be used as a foreign key in another table to form a link/relationship between the tables By example: Identification of a primary key in a table Describing <u>that</u> primary key in another table as a foreign key	2
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9608_s19_qp_11_sc_2.c.i

QUESTION

(c) The database has the following tables:

```
CUSTOMER (CustomerID, CompanyName)
SOFTWARE (SoftwareID, SoftwareName, OperatingSystem, Description)
LICENCE (LicenceID, CustomerID, SoftwareID, DateOfPurchase,
         LicenceType, Cost, ExpiryDate)
```

(i) Identify the type of relationship that exists between the tables CUSTOMER and LICENCE.

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..... [1]

MARK SCHEME

2(c)(i)	1-to-many // 1 customer to/has many licences	1
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9608_s19_qp_11_sc_2.c.ii

QUESTION

(ii) Describe how the relationship is created between the tables CUSTOMER and LICENCE.

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..... [2]

MARK SCHEME

2(c)(ii)	1 mark per bullet point <u>CustomerID</u> is the Primary key in <u>CUSTOMER</u> table Links to <u>CustomerID</u> as a Foreign key in <u>LICENCE</u> table	2
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9608_w18_qp_11_sc_7.b

QUESTION

(b) Explain how primary and foreign keys are used to link the tables in the movie theatre database.

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.....[4]

MARK SCHEME

7(b)	1 mark per bullet point MovieID is the Primary Key in MOVIE ... <u>links to</u> MovieID which is the Foreign Key in MOVIESCHEDULE ScreenNumber is the Primary Key in SCREEN ... <u>links to</u> ScreenNumber which is the Foreign Key in MOVIESCHEDULE	4
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QUESTION

- 7 A social media website has a relational database, `WEBDATA`, that stores the site's information.

The database has three tables to store users' details, and details of the images and text that they post.

`USER`(`UserName`, `FirstName`, `SecondName`, `DateOfBirth`)

`PHOTO`(`PhotoID`, `UserName`, `Comment`, `UploadDate`)

`TEXTPOST`(`PostID`, `UserName`, `DateOfPost`, `TheText`)

- (a) (i) Explain how the relationship between the tables `USER` and `PHOTO` has been implemented.

.....

[2]

MARK SCHEME

7(a)(i)	1 mark per bullet <u><code>UserName</code></u> is the primary key in <u><code>USER</code></u> <u><code>UserName</code></u> is (included as) a foreign key in <u><code>PHOTO</code></u>	2
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QUESTION

- (b) A database administrator decides to enforce referential integrity.

Use an example from the database `WEBDATA` to explain what is meant by **referential integrity**.

.....

[3]

MARK SCHEME

7(b)	1 mark per bullet to max 2 for explanation Referential integrity is making sure tables do not try to reference data which does not exist // A value of one attribute of a table exists as a value of another attribute in a different table A primary key cannot be deleted unless all dependent records are already deleted Cascading delete A primary key cannot be updated unless all dependent records are already updated Cascading update / edit Every foreign key value has a matching value in the corresponding primary key The foreign keys must be the same data type as the corresponding primary key 1 mark for a suitable example e.g. A <code>UserName</code> cannot be deleted from the <code>USER</code> table if they have a related <code>photo/textpost</code> If <code>UserName</code> is updated in <code>USER</code> table, it must also be updated in <code>PHOTO</code> and <code>TEXTPOST</code> tables Cannot create/edit a record in <code>TEXTPOST</code> / <code>PHOTO</code> without a matching entry in <code>USER</code> table	3
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9608_w17_qp_13_sc_7.a.i

QUESTION

- 7 A clinic is staffed by several doctors. The clinic serves thousands of patients. Each day and at any one time, there is only one doctor in the clinic available for appointments.

The clinic stores patient, doctor and appointment data in a relational database.

- (a) (i) Underline the primary key for each table in the following suggested table designs.

PATIENT(PatientID, PatientName, Address, Gender)

DOCTOR(DoctorID, Gender, Qualification)

APPOINTMENT(AppointmentDate, AppointmentTime, DoctorID, PatientID)

[2]

MARK SCHEME

7(a)(i)	<u>PatientID</u> <u>DoctorID</u> } (1) <u>AppointmentDate, AppointmentTime</u> (1)	2
---------	---	---

QUESTION

- (b) The doctors are concerned that many patients make appointments but do not attend them.

Describe the changes to the table designs that could be made to store this information.

.....
[2]

MARK SCHEME

7(b)	Two marks from: Either: Add an attribute (for example <i>Attended</i>) To the appointment table // <i>APPOINTMENT</i> Or: Add an attribute (for example <i>AppointmentsMissed</i>) To the patient table // <i>PATIENT</i>	2
------	--	----------

QUESTION

- (c) The doctors are about to set up a new clinic in the neighbouring village, SITE-B.

The original location is identified as SITE-A.

A new table is designed to store the ID of the doctor who is able to work at each site.

DOCTOR-AVAILABILITY (DoctorID, *Site*)

Five entries stored in the table are:

DoctorID	Site
098	SITE-A
074	SITE-A
117	SITE-B
098	SITE-B
033	SITE-B

- (i) State what this data shows about the availability of the doctor with the ID of 098.

.....
[1]

MARK SCHEME

7(c)(i)	Available to work at both <i>SITE-A</i> and <i>SITE-B</i>	1
---------	---	----------

QUESTION

(ii) Describe why this update could cause problems with the existing data stored.

.....
.....
.....[2]

MARK SCHEME

7(d)(ii)	1 Mark per bullet, max 2	Max 2
	Referential integrity should be maintained // Referential integrity could be violated. Data becomes inconsistent There may be records in the <code>APPOINTMENT</code> table showing doctor ID 117 The <code>APPOINTMENT</code> table might not be automatically updated Records in the <code>APPOINTMENT</code> table will become orphaned	

QUESTION

7 A company takes customer service for its clients very seriously.

The client

- The client names are unique.

A visit

- The company arranges a date for a visit to gather feedback from a client.
- A visit to a client never takes more than one day.
- Over time, the client receives many visits.

Staff (Interviewers)

- One or more staff attend the visit.
- If there is more than one staff member visiting, each performs a separate interview.

Interviews

- Each interview is classified as either 'general' or by some specialism, for example, marketing, customer service or sales.
- A report is produced for each interview, *InterviewText*.
- Each interview is conducted by a single staff member.

The client, visit, staff and interview data will be stored in a relational database.

(a) (i) Underline the primary key for each table in the following suggested table designs.

STAFF(StaffID, StaffName, Department)

CLIENT(ClientName, Address, Town)

VISIT(ClientName, VisitDate)

INTERVIEW(ClientName, VisitDate, StaffID, SpecialistFocus, InterviewText)

[3]

MARK SCHEME

7(a)(i)	1 Mark for correct primary key identified in both STAFF and CLIENT STAFF(<u>StaffID</u>, StaffName, Department) CLIENT(<u>ClientName</u>, Address, Town) 1 Mark for correct primary key identified in VISIT VISIT(<u>ClientName</u>, VisitDate) 1 Mark for correct primary key identified in INTERVIEW INTERVIEW(<u>ClientName</u>, VisitDate, StaffID, SpecialistFocus, InterviewText)	3
---------	---	---

QUESTION

(ii) For each of the pairs of entities, A, B and C, draw the relationship between the two entities.

A

CLIENT

VISIT

B

VISIT

INTERVIEW

C

INTERVIEW

STAFF

[3]

MARK SCHEME

7(a)(ii)	1 Mark for each correct relationship CLIENT VISIT VISIT INTERVIEW INTERVIEW STAFF	3
----------	--	---

QUESTION

(b) The company decides to produce a visit report, `VisitReportText`, for each visit made.

This text will be produced from the one or more interview texts obtained at the visit.

State how one or more of the given table designs can be changed to add this attribute.

.....

.....[1]

MARK SCHEME

7(b)	1 Mark for correct answer Add attribute <code>VisitReportText</code> to table <u>VISIT</u>	1
------	--	---

QUESTION

- (ii) Describe why this update could cause problems with the existing data stored.

.....

[2]

MARK SCHEME

7(c)(ii)	1 Mark per bullet, max 2 Referential integrity should be maintained // Referential integrity could be violated Data becomes inconsistent There may be records in the <code>VISIT</code> and <code>INTERVIEW</code> tables / other tables with client name ABC Holdings The <code>ClientName</code> in the <code>VISIT</code> and <code>INTERVIEW</code> tables / other tables might not be automatically updated Records in the <code>VISIT</code> and <code>INTERVIEW</code> tables / other tables will become orphaned	2
----------	--	----------

QUESTION

- 1 A hospital is divided into two areas, Area A and Area B. Each area has several wards. All the ward names are different.

A number of nurses are based in Area A. These nurses always work on the same ward. Each nurse has a unique Nurse ID of `STRING` data type.



- (a) Describe the relationship shown above.

.....
[1]

MARK SCHEME

1(a)	Many-to-one	1
------	-------------	----------

9608_s17_qp_13_sc_1.b.i

QUESTION

- (b) A relational database is created to store the ward and nurse data. The two table designs for Area A are:

A-WARD (WardName, NumberOfBeds)

A-NURSE (NurseID, FirstName, FamilyName,)

- (i) Complete the design for the A-NURSE table. [1]

MARK SCHEME

1(b)(i)	A-NURSE (<u>NurseID</u> , FirstName, FamilyName, WardName)	1
---------	---	----------

9608_s17_qp_13_sc_1.b.ii

QUESTION

- (ii) Explain how the relationship in **part (a)** is implemented.

.....
.....
.....
.....[2]

MARK SCHEME

1(b)(ii)	The primary key <u>WardName</u> in the A-WARD table ...	1	2
	... links to the foreign key <u>WardName</u> in the A-NURSE table.	1	

QUESTION

(c) In Area B of the hospital, there are a number of wards and a number of nurses.

Each Area B ward has a specialism.

Each Area B nurse has a specialism.

A nurse can be asked to work in any of the Area B wards where their specialism matches with the ward specialism.

The relationship for Area B of the hospital is:



(i) Explain what the degree of relationship is between the entities B-NURSE and B-WARD.

.....
[1]

MARK SCHEME

1(c)(i)	Many-to-many relationship	1
---------	---------------------------	---

QUESTION

(ii) The design for the Area B data is as follows:

B-NURSE (NurseID, FirstName, FamilyName, Specialism)

B-WARD (WardName, NumberOfBeds, Specialism)

B-WARD-NURSE (.....)

Complete the attributes for the third table. Underline its primary key. [2]

MARK SCHEME

1(c)(ii)	B-WARD-NURSE (<u>WardName</u> , <u>NurseID</u>)	2
	Both attributes (with no additions)	1
	Joint primary key correctly underlined	1

QUESTION

- 1 Some shops belong to the Rainbow Retail buying group. They buy their goods from one or more suppliers.

Each shop has:

- a unique shop ID
- a single retail specialism (for example, food, electrical, garden).

Each supplier has:

- a unique supplier ID
- a similar single specialism recorded.

Rainbow Retail creates a relational database to record data about the shops and their suppliers.

The entity-relationship (E-R) diagram for the relationship between the `SHOP` and `SUPPLIER` tables is shown.



- (a) Explain what the degree of relationship is between the entities `SHOP` and `SUPPLIER`.

.....
.....[1]

MARK SCHEME

1(a)	Many-to-many relationship	1
------	---------------------------	---

QUESTION

(ii) Complete the following table to show for each database table:

- the primary key
- the foreign key(s) (if any):
 - Each table may contain none, one or more foreign key(s).
 - For a table with no foreign key, write 'None'.
- an explanation for the use of any foreign key.

Table	Primary key	Foreign key(s) (if any)	Explanation
SHOP			
SUPPLIER			
SHOP-SUPPLIER			

[5]

MARK SCHEME

1(b)(ii)

Table	Primary key	Foreign keys(s) (if any)	Explanation
SHOP	ShopID	None	
SUPPLIER	SupplierID	None	
SHOP-SUPPLIER	ShopID AND SupplierID	ShopID OR SupplierID (or both)	To create a link with the SHOP or SUPPLIER table.

SHOP has primary key ShopID and SUPPLIER has primary key SupplierID

SHOP-SUPPLIER has primary key ShopID + SupplierID

Both SHOP and SUPPLIER show foreign key as 'None'

SHOP-SUPPLIER shows foreign key ShopID or SupplierID

Explanation for SHOP-SUPPLIER foreign key describes ShopID or SupplierID creating a link

5

1

1

1

1

1

9608_s17_qp_12_sc_1.b.iii

QUESTION

- (iii) The database designer has implemented `SUPPLIER.ContactPerson` as a secondary key.

Describe the reason for this.

.....

.....

.....

.....[2]

MARK SCHEME

1(b)(iii)	Two from: The database user will <u>frequently</u> want to search on contact name The contact name attribute has been indexed It allows for a <u>fast/faster</u> search using contact name	1 1 1	Max 2
-----------	--	-------------	--------------

9608_s17_qp_11_sc_1.a

QUESTION

- 1 A hospital is divided into two areas, Area A and Area B. Each area has several wards. All the ward names are different.

A number of nurses are based in Area A. These nurses always work on the same ward. Each nurse has a unique Nurse ID of `STRING` data type.



- (a) Describe the relationship shown above.

.....

.....[1]

MARK SCHEME

1(a)	Many-to-one	1
------	-------------	----------

9608_s17_qp_11_sc_1.b.i

QUESTION

(b) A relational database is created to store the ward and nurse data. The two table designs for Area A are:

A-WARD (WardName, NumberOfBeds)

A-NURSE (NurseID, FirstName, FamilyName,)

(i) Complete the design for the A-NURSE table. [1]

MARK SCHEME

1(b)(i)	A-NURSE (<u>NurseID</u> , FirstName, FamilyName, WardName)	1
---------	---	---

9608_s17_qp_11_sc_1.b.ii

QUESTION

(ii) Explain how the relationship in **part (a)** is implemented.

.....

.....

.....

.....[2]

MARK SCHEME

1(b)(ii)	The primary key <u>WardName</u> in the A-WARD table ...	1	2
	... links to the foreign key <u>WardName</u> in the A-NURSE table.	1	

QUESTION

(c) In Area B of the hospital, there are a number of wards and a number of nurses.

Each Area B ward has a specialism.

Each Area B nurse has a specialism.

A nurse can be asked to work in any of the Area B wards where their specialism matches with the ward specialism.

The relationship for Area B of the hospital is:

B-NURSE

B-WARD

(i) Explain what the degree of relationship is between the entities B-NURSE and B-WARD.

.....

.....[1]

MARK SCHEME

1(c)(i)	Many-to-many relationship	1
---------	---------------------------	---

QUESTION

(ii) The design for the Area B data is as follows:

B-NURSE (NurseID, FirstName, FamilyName, Specialism)

B-WARD (WardName, NumberOfBeds, Specialism)

B-WARD-NURSE (.....)

Complete the attributes for the third table. Underline its primary key. [2]

MARK SCHEME

1(c)(ii)	B-WARD-NURSE (<u>WardName</u> , <u>NurseID</u>)	2
	Both attributes (with no additions)	1
	Joint primary key correctly underlined	1

QUESTION

- 1 (a) Five descriptions and seven relational database terms are shown below.

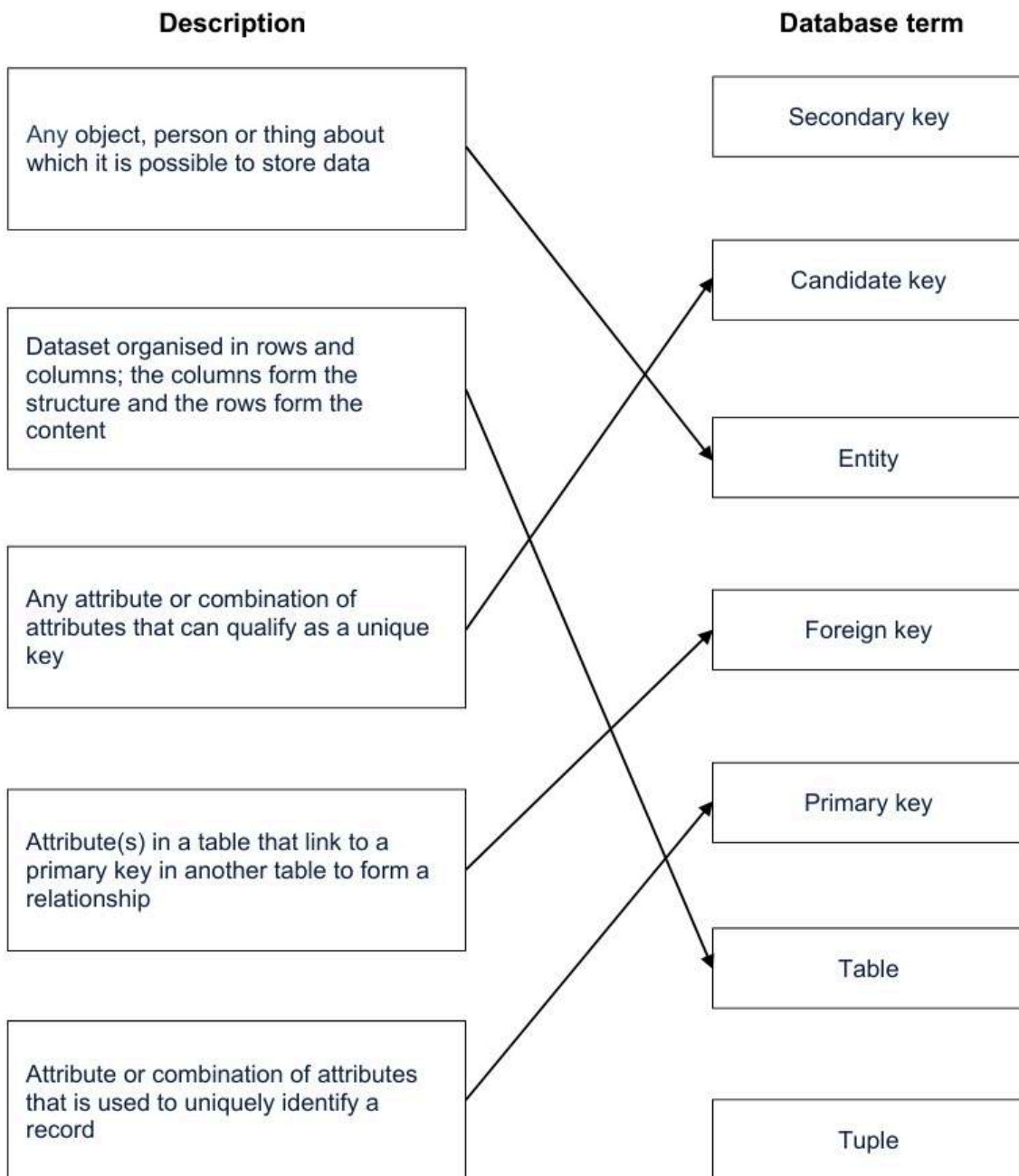
Draw a line to link each description to its correct database term.

Description	Database term
Any object, person or thing about which it is possible to store data	Secondary key
Dataset organised in rows and columns; the columns form the structure and the rows form the content	Candidate key
Any attribute or combination of attributes that can act as a unique key	Entity
Attribute(s) in a table that link to the primary key in another table to form a relationship	Foreign key
Attribute or combination of attributes that is used to uniquely identify a record	Primary key
	Table
	Tuple

[5]

MARK SCHEME

- 1 (a) One mark** for each correct line.
Two lines from any box on left means no mark for that description.



[5]

QUESTION

(b) Explain what is meant by referential integrity.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

(b) Any **three** from:

- Ensures related data in tables are consistent
- If one table has a foreign key (the 'foreign' table)...
- ... then it is not possible to add a record to that table / the 'foreign' table
- ... unless there is a corresponding record in the linked table with a corresponding primary key (the 'primary' table)
- Cascading delete
- If a record is deleted in the 'primary' table...
- all corresponding linked records in 'foreign' tables must also be deleted
- Cascading update
- If a record in the 'primary' table is modified...
- ... all linked records in foreign tables will also be modified

[3]

QUESTION

- 1 (a) Five descriptions and seven relational database terms are shown below.

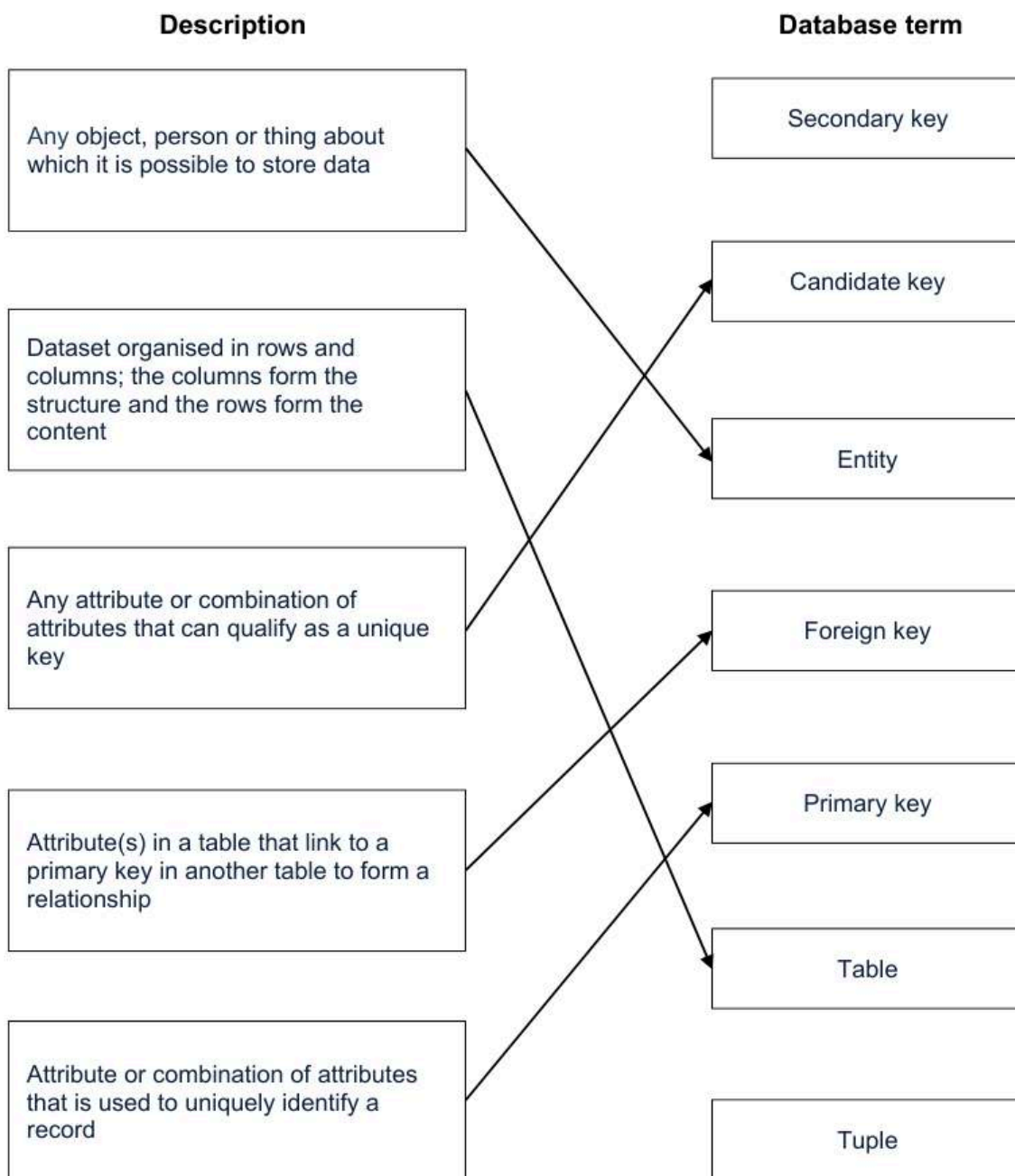
Draw a line to link each description to its correct database term.

Description	Database term
Any object, person or thing about which it is possible to store data	Secondary key
Dataset organised in rows and columns; the columns form the structure and the rows form the content	Candidate key
Any attribute or combination of attributes that can act as a unique key	Entity
Attribute(s) in a table that link to the primary key in another table to form a relationship	Foreign key
Attribute or combination of attributes that is used to uniquely identify a record	Primary key
	Table
	Tuple

[5]

MARK SCHEME

- 1 (a) One mark** for each correct line.
Two lines from any box on left means no mark for that description.



[5]

QUESTION

(b) Explain what is meant by referential integrity.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

(b) Any **three** from:

- Ensures related data in tables are consistent
- If one table has a foreign key (the 'foreign' table)...
- ... then it is not possible to add a record to that table / the 'foreign' table
- ... unless there is a corresponding record in the linked table with a corresponding primary key (the 'primary' table)
- Cascading delete
- If a record is deleted in the 'primary' table...
- all corresponding linked records in 'foreign' tables must also be deleted
- Cascading update
- If a record in the 'primary' table is modified...
- ... all linked records in foreign tables will also be modified

[3]

QUESTION

(ii) State the type of relationship that exists between STUDENT and STUDENT-QUALIFICATION.

.....[1]

MARK SCHEME

(ii) **One-to-many**

[1]

9608_s16_qp_13_sc_5.c.iii

QUESTION

- (iii) Describe how the relationship between `QUALIFICATION` and `STUDENT-QUALIFICATION` is implemented.

.....
.....
.....
.....[2]

MARK SCHEME

- (iii) **Two** points from:

[2]

- The primary key in the `QUALIFICATION` table is `QualCode`.
- The foreign key in the `STUDENT-QUALIFICATION` table is `QualCode`.
- The primary key of `QUALIFICATION` is also included in `QualCode`.

9608_s16_qp_12_sc_8.b.i

QUESTION

- (b) The database design has three tables to store the classes that students attend.

`STUDENT` (`StudentID`, `FirstName`, `LastName`, `Year`, `TutorGroup`)

`CLASS` (`ClassID`, `Subject`)

`CLASS-GROUP` (`StudentID`, `ClassID`)

Primary keys are not shown.

There is a one-to-many relationship between `CLASS` and `CLASS-GROUP`.

- (i) Describe how this relationship is implemented.

.....
.....
.....[2]

MARK SCHEME

(b) (i) Two points from:

[2]

- The Primary Key in **CLASS** is ClassID
- The Foreign Key of **CLASS-GROUP** is ClassID.
- The Primary Key of **CLASS** is also included in **CLASS-GROUP** as a Foreign Key, (which links to **CLASS** table)

9608_s16_qp_12_sc_8.b.ii

QUESTION

(ii) Describe the relationship between **CLASS-GROUP** and **STUDENT**.

.....[1]

MARK SCHEME

(ii) Many-to-one

[1]

9608_s16_qp_11_sc_8.b.i

QUESTION

(b) The database design has three tables to store the classes that students attend.

STUDENT(StudentID, FirstName, LastName, Year, TutorGroup)

CLASS(ClassID, Subject)

CLASS-GROUP(StudentID, ClassID)

Primary keys are not shown.

There is a one-to-many relationship between **CLASS** and **CLASS-GROUP**.

(i) Describe how this relationship is implemented.

.....
.....
.....[2]

MARK SCHEME

(b) (i) Two points from: [2]

- The Primary Key in **CLASS** is ClassID
- The Foreign Key of **CLASS-GROUP** is ClassID.
- The Primary Key of **CLASS** is also included in **CLASS-GROUP** as a Foreign Key, (which links to **CLASS** table)

9608_s16_qp_11_sc_8.b.ii

QUESTION

(ii) Describe the relationship between **CLASS-GROUP** and **STUDENT**.

.....[1]

MARK SCHEME

(ii) Many-to-one [1]

9608_s15_qp_12_sc_9.c.i

QUESTION

(c) (i) A relationship between the two tables has been implemented.

Explain how this has been done.

.....
.....
.....
.....
.....[2]

MARK SCHEME

(c) (i) Any **two** from:

- primary key of SalesPerson table is FirstName
- links to FirstName in SalesProducts table
- FirstName in SalesProductsS table is foreign key

[2]

QUESTION

(c) (i) A relationship between the two tables has been implemented.

Explain how this has been done.

.....

.....

.....

.....

.....[2]

MARK SCHEME

(c) (i) Any **two** from:

- primary key of SalesPerson table is FirstName
- links to FirstName in SalesProducts table
- FirstName in SalesProductsS table is foreign key

[2]

Use an entity-relationship (E-R) diagram to document a database design

9608_w21_qp_11_sc_9.a

QUESTION

- 9 An airline company uses a relational database to store data about passengers and flights.

Part of the database is shown.

PASSENGER(PassengerID, FirstName, LastName)

FLIGHT(FlightID, FlightDate, FlightTime)

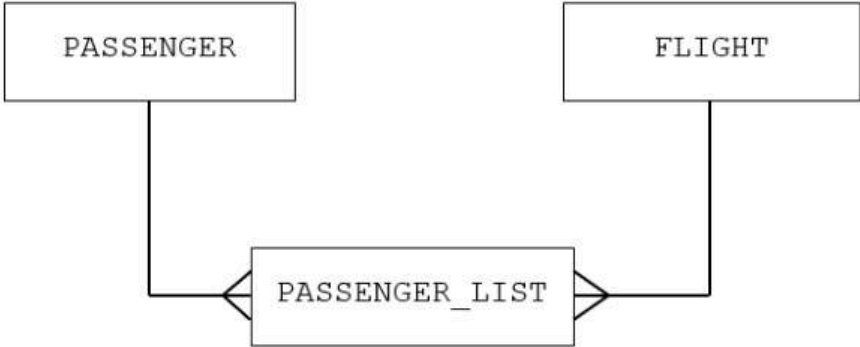
PASSENGER_LIST(FlightID, PassengerID, SeatNo)

- (a) Complete the entity-relationship (E-R) diagram to show the relationships between the given tables.



[2]

MARK SCHEME

9(a)	One mark per correct relation	2
 <pre> graph TD PASSENGER[PASSENGER] ===> PASSENGER_LIST[PASSENGER_LIST] FLIGHT[FLIGHT] ===> PASSENGER_LIST style PASSENGER fill:#fff,stroke:#000 style FLIGHT fill:#fff,stroke:#000 style PASSENGER_LIST fill:#fff,stroke:#000 </pre>		

9608_s21_qp_13_sc_8.a

QUESTION

- 8 A charity has a relational database, CHARITY_PROJECT.

The database has three tables to store information about the charity's employees, the projects and the project teams.

EMPLOYEE(EmployeeID, FirstName, LastName, Salary, DOB)

PROJECT(ProjectID, Name, Location, ProjectStartDate)

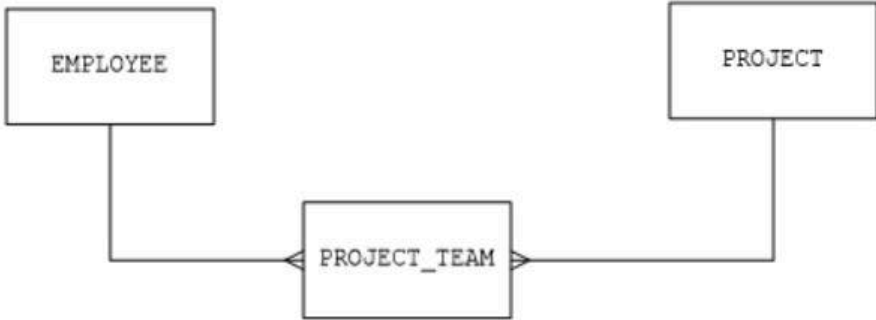
PROJECT_TEAM(EmployeeID, ProjectID, JoinedProjectDate)

- (a) Complete the entity relationship (E-R) diagram to show the relationships between these tables.



[2]

MARK SCHEME

8(a)	1 mark for each correct join 	2
------	---	----------

9608_s21_qp_11_sc_5.a

QUESTION

- 5 A company runs activity courses. It is creating a relational database to store details of the courses it runs.

The database has five tables:

EMPLOYEE(EmployeeID, FirstName, LastName, Role, Language)

BOOKING(BookingID, CustomerID, CourseID)

CUSTOMER(CustomerID, FirstName, LastName)

COURSE(CourseID, Title, Level, Date)

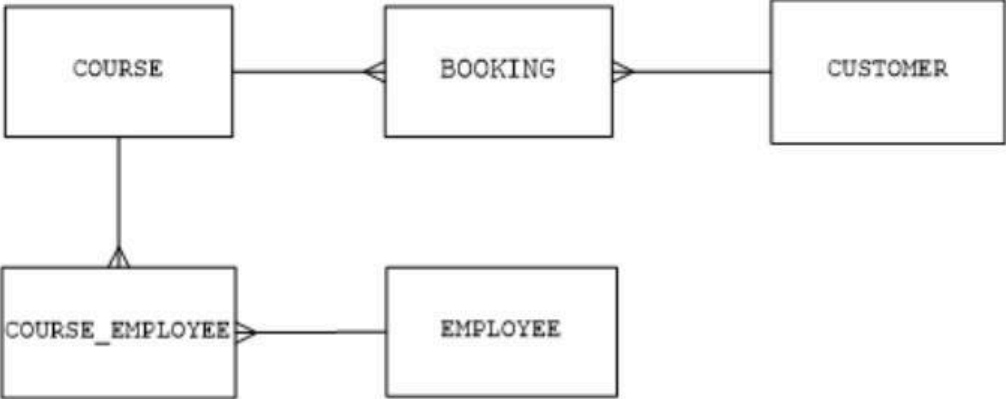
COURSE_EMPLOYEE(CourseID, EmployeeID)

- (a) Complete the entity relationship (E-R) diagram for the database.



[4]

MARK SCHEME

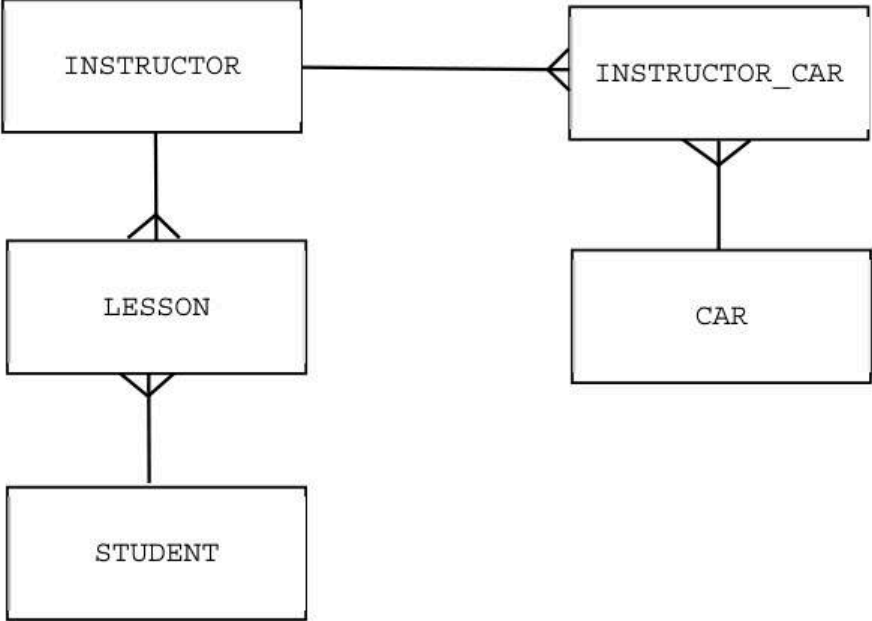
5(a)	1 mark for each link: • 1 CUSTOMER to many BOOKING • 1 COURSE to many BOOKING • 1 COURSE to many COURSE_EMPLOYEE • 1 EMPLOYEE to many COURSE_EMPLOYEE 	4
------	--	---

9608_s20_qp_11_sc_7.b

QUESTION

(b) Complete the entity-relationship diagram for the database DRIVING_SCHOOL. INSTRUCTOR LESSON STUDENT INSTRUCTOR_CAR CAR	
[4]	

MARK SCHEME

7(b)	1 mark for each correct link  <pre> graph TD INSTRUCTOR[INSTRUCTOR] --> INSTRUCTOR_CAR[INSTRUCTOR_CAR] LESSON[LESSON] --> STUDENT[STUDENT] INSTRUCTOR_CAR --> CAR[CAR] </pre>	4
------	--	---

9608_w19_qp_12_sc_4.b.i

QUESTION

(b) The **normalised** relational database, `SPORTS_CLUB`, has the following table design.

`MEMBER`(MemberID, FirstName, LastName, MembershipType)

`SESSION`(SessionID, Description, SessionDate, SessionTime, NumberMembers)

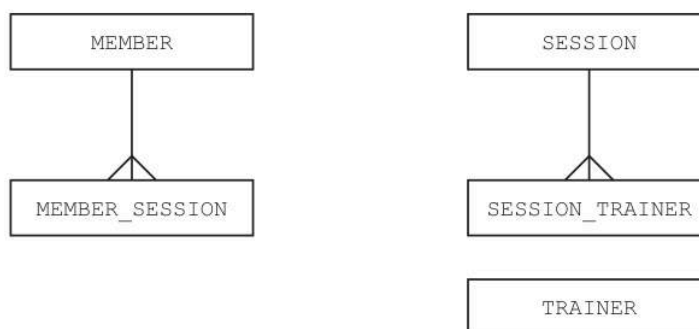
`TRAINER`(TrainerID, TrainerFirstName, TrainerLastName)

`MEMBER_SESSION`(MemberID, SessionID)

`SESSION_TRAINER`(SessionID, TrainerID)

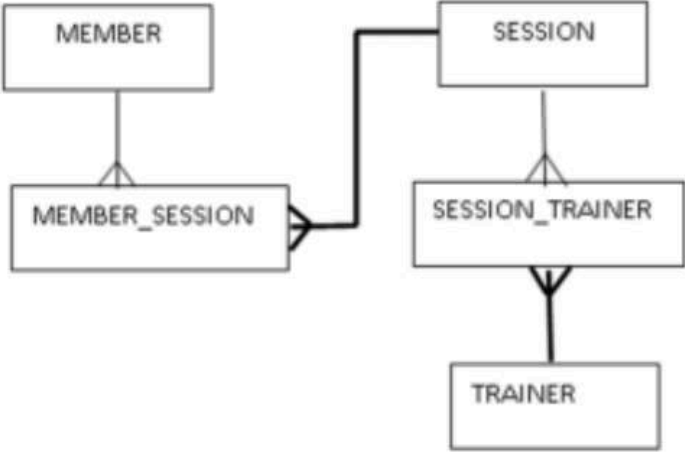
(i) Anushka has designed an entity-relationship (E-R) diagram for `SPORTS_CLUB`.

Complete the entity-relationship (E-R) diagram.



[2]

MARK SCHEME

4(b)(i)	1 mark for SESSION to MEMBER_SESSION (one-to-many) 1 mark for TRAINER to SESSION_TRAINER (one-to-many)  <pre> graph TD MEMBER[MEMBER] --> MEMBER_SESSION[MEMBER_SESSION] SESSION[SESSION] --> MEMBER_SESSION SESSION --> SESSION_TRAINER[SESSION_TRAINER] TRAINER[TRAINER] --> SESSION_TRAINER </pre>	2
---------	--	---

9608_s19_qp_13_sc_3.c

QUESTION

(c) In the company:

- An employee can be a manager.
- A department can have several managers and several employees.
- An employee can only belong to one department.

The EMPLOYEES database has three tables:

EMPLOYEE_DATA (EmployeeID, FirstName, LastName, DateOfBirth, Gender, DepartmentNumber)

DEPARTMENT (DepartmentNumber, DepartmentName)

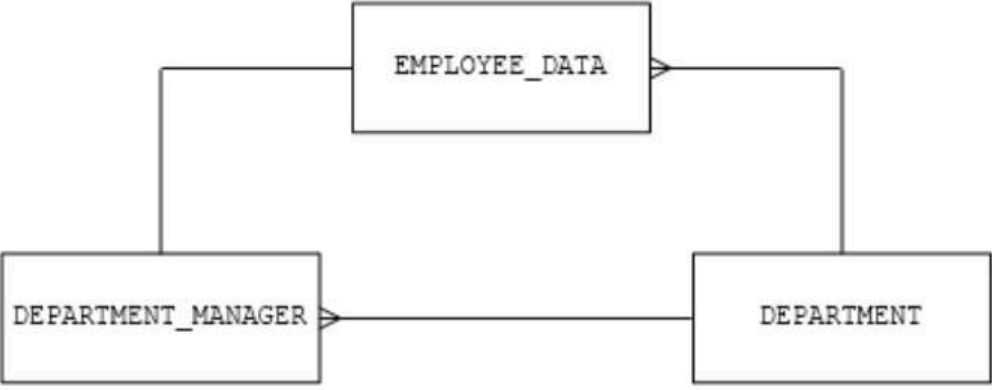
DEPARTMENT_MANAGER (DepartmentNumber, EmployeeID, role)

Complete the entity-relationship (E-R) diagram for the EMPLOYEES database.



[3]

MARK SCHEME

3(c)	1 mark for each correct link  <pre> graph TD EMPLOYEE_DATA[EMPLOYEE_DATA] --- DEPARTMENT_MANAGER[DEPARTMENT_MANAGER] EMPLOYEE_DATA --- DEPARTMENT[DEPARTMENT] DEPARTMENT_MANAGER --- DEPARTMENT </pre>	3
------	--	----------

9608_w18_qp_11_sc_7.a

QUESTION

- 7 A movie theatre has a relational database that stores the movie schedule, and information about the movies. The theatre has several screens that play movies at the same time.

The database has three tables to store information about the movies, the screens and the movie schedule.

MOVIE (MovieID, Title, Length, Rating)

SCREEN (ScreenNumber, NumberSeats)

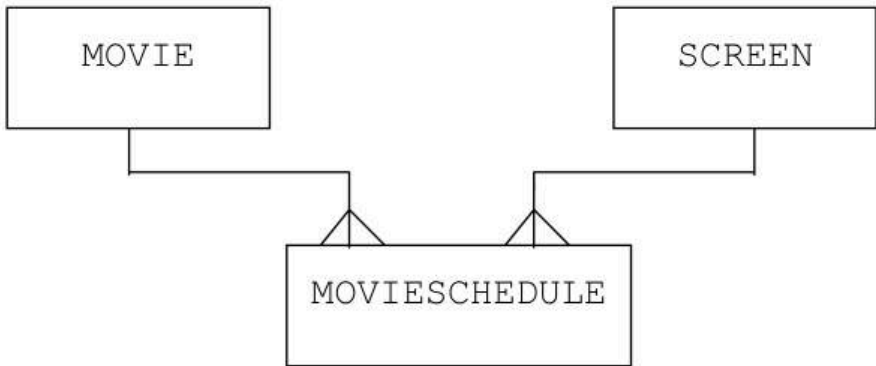
MOVIESCHEDULE (ScheduleID, MovieID, ScreenNumber, Time)

- (a) Complete the entity-relationship (E-R) diagram to show the relationships between these tables.



[2]

MARK SCHEME

7(a)	1 mark for each correct join 	2
------	--	---

9608_s18_qp_13_sc_2.a

QUESTION

- 2 A company writes applications (apps) for smartphones. The company has a relational database, PURPLEGAME, which stores the information for one of its online game apps.

The database has three tables to store player's details, dates when they have logged into the app and in-app purchase details.

LOGIN(LoginID, PlayerID, Date)

PURCHASE(PurchaseID, PlayerID, PurchaseDate, Cost)

PLAYER(PlayerID, PlayerName, SkillLevel)

- (a) Draw the entity-relationship (E-R) diagram to show the relationships between the three tables.

[2]

MARK SCHEME

2(a)	1 mark for each correct relationship 	2
------	--	---

9608_s18_qp_12_sc_7.a.ii

QUESTION

- (ii) Draw the entity-relationship (E-R) diagram to show the relationships between the three tables.

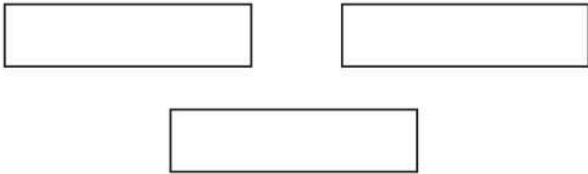
[2]

MARK SCHEME

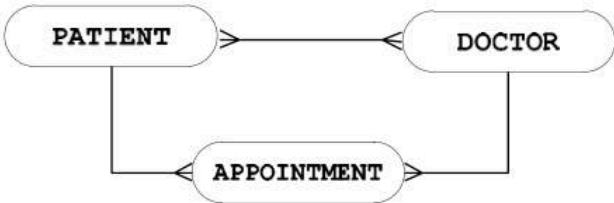
7(a)(ii)	1 mark for each correct relationship 	2
----------	---	----------

9608_w17_qp_13_sc_7.a.ii

QUESTION

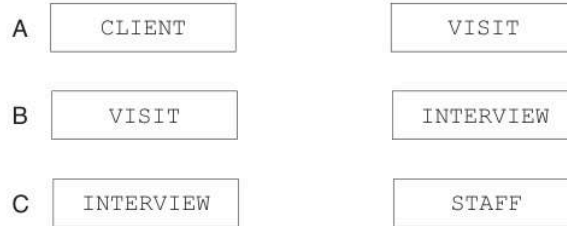
(ii) Complete the following entity-relationship (E-R) diagram for this design. 		
		[2]

MARK SCHEME

7(a)(ii)	 One PATIENT attends many APPOINTMENTs One DOCTOR takes many APPOINTMENTs Special case for 1 mark only (only if no one to many relationships shown) Many PATIENTs are seen by many DOCTORs	2
----------	--	----------

QUESTION

- (ii) For each of the pairs of entities, A, B and C, draw the relationship between the two entities.



[3]

MARK SCHEME

7(a)(ii)	1 Mark for each correct relationship	3
----------	--	----------

QUESTION

- 1 A hospital is divided into two areas, Area A and Area B. Each area has several wards. All the ward names are different.

A number of nurses are based in Area A. These nurses always work on the same ward. Each nurse has a unique Nurse ID of `STRING` data type.



- (a) Describe the relationship shown above.

.....
[1]

MARK SCHEME

1(a)	Many-to-one	1
------	-------------	----------

QUESTION

(iii) Draw the relationships on the entity-relationship (E-R) diagram.



[2]

MARK SCHEME

1(c)(iii)	Correct relationship between B-NURSE and B-WARD-NURSE Correct relationship between B-WARD and B-WARD-NURSE	2
		1
		1

QUESTION

- 1 Some shops belong to the Rainbow Retail buying group. They buy their goods from one or more suppliers.

Each shop has:

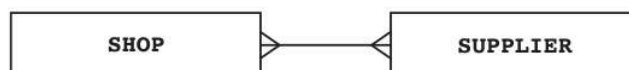
- a unique shop ID
- a single retail specialism (for example, food, electrical, garden).

Each supplier has:

- a unique supplier ID
- a similar single specialism recorded.

Rainbow Retail creates a relational database to record data about the shops and their suppliers.

The entity-relationship (E-R) diagram for the relationship between the **SHOP** and **SUPPLIER** tables is shown.



- (a) Explain what the degree of relationship is between the entities **SHOP** and **SUPPLIER**.

.....

.....[1]

MARK SCHEME

1(a)	Many-to-many relationship	1
------	---------------------------	---

9608_s17_qp_12_sc_1.b.i

QUESTION

The database design is as follows:

SHOP(ShopID, ShopName, Location, RetailSpecialism)

SUPPLIER(SupplierID, SupplierName, ContactPerson, RetailSpecialism)

SHOP-SUPPLIER(ShopID, SupplierID)

The SHOP-SUPPLIER table stores the suppliers that each shop has previously used.

Primary keys are not shown.

(b) (i) Label the entities and draw the relationships to complete the revised E-R diagram.

SUPPLIER

[3]

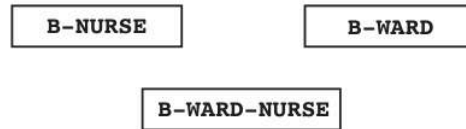
MARK SCHEME

1(b)(i)	<pre>graph TD; SHOP([SHOP]) --- SHOP_SUPPLIER([SHOP-SUPPLIER]); SUPPLIER([SUPPLIER]) --- SHOP_SUPPLIER</pre> Both entities correctly labelled1 Correct relationship between SHOP and SHOP-SUPPLIER1 Correct relationship between SUPPLIER and SHOP-SUPPLIER1	3
---------	--	---

9608_s17_qp_11_sc_1.c.iii

QUESTION

(iii) Draw the relationships on the entity-relationship (E-R) diagram.



[2]

MARK SCHEME

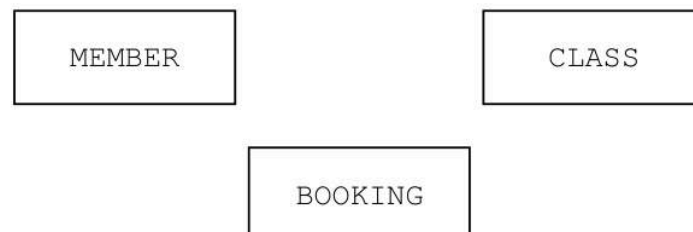
1(c)(iii)	<pre> graph TD B_NURSE((B-NURSE)) --- B_WARD_NURSE((B-WARD-NURSE)) B_WARD((B-WARD)) --- B_WARD_NURSE </pre> Correct relationship between B-NURSE and B-WARD-NURSE Correct relationship between B-WARD and B-WARD-NURSE	2
		1
		1

9608_w16_qp_12_sc_9.b

QUESTION

(b) The programmer decides to use three tables: MEMBER, BOOKING and CLASS.

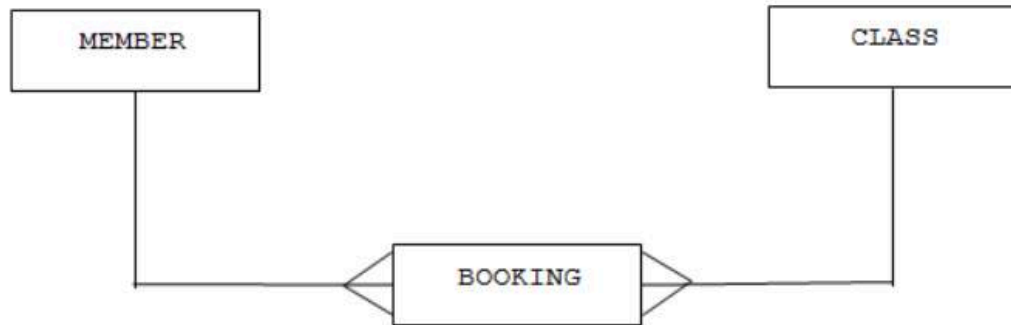
Complete the Entity-Relationship (E-R) diagram to show the relationships between these tables.



[2]

MARK SCHEME

(b) **ONE** mark for each correct relationship as shown.



[2]

QUESTION

- (c) The database design has three tables to store the qualifications and grades each student has attained. The following is a sample of the data from each table.

STUDENT

StudentID	FirstName	LastName	Tutor
001AT	Ahmad	Tan	11A
003JL	Jane	Li	11B
011HJ	Heather	Jones	10A

QUALIFICATION

QualCode	Level	Subject
CS1	IGCSE	Computer Science
MT9	IGCSE	Maths
SC12	IGCSE	Science

STUDENT-QUALIFICATION

QualCode	StudentID	Grade	DateOfAward
SC12	011HJ	A	31/8/2014
SC12	003JL	C	31/8/2014
CS1	003JL	B	31/8/2014

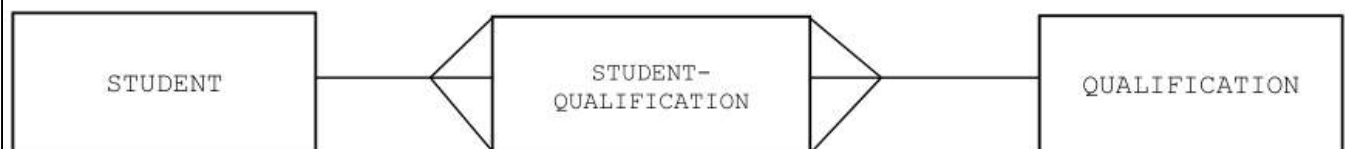
- (i) Draw an Entity-Relationship (E-R) diagram to show the relationships between these three tables.

[2]

MARK SCHEME

- (c) (i) One mark for each correct relationship.

[2]



Show understanding of the normalisation process

9608_s21_qp_12_sc_4.b

QUESTION

- (b) Each course has a destination and a trip date, and is run by a guide. Customers can book a place on one or more courses.

The following table shows part of the stored file:

Customer name	Customer date of birth	Guide	Destination	Trip date
Jay Patel	10/10/1976	Artem	Elbrus	06/03/2023
Paul Schloss	10/04/1999	Kenton	K2	01/03/2022
Mina Wang	03/03/2000	Kenton	K2	01/03/2022
Paul Schloss	10/04/1999	Safia	Mont Blanc	07/07/2024
Jay Patel	10/10/1976	Safia	K2	04/04/2023

The company has decided to create a relational database to replace the current file-based approach.

Identify **three** reasons why the data in this table is not in First Normal Form (1NF).

- 1
.....
- 2
.....
- 3
.....

[3]

MARK SCHEME

4(b)	1 mark per bullet point (max 3) - There is no unique <u>primary</u> key - Customer name is not atomic // Customer name needs to be split into first name and last name // by example - Customer name / Customer date of birth / Destination / Guide name / Trip have repeated groups of attributes	3
------	--	----------

QUESTION

- (ii) The relational database design needs to be normalised. The following statements describe the three stages of database normalisation.

Complete the statements by filling in the missing words.

For a database to be in First Normal Form (1NF) there must be no groups of attributes.

For a database to be in Second Normal Form (2NF), it must be in 1NF, and contain no key dependencies.

For a database to be in Third Normal Form (3NF), it must be in 2NF, and all attributes must be fully dependent on the

[4]

MARK SCHEME

4(a)(ii)	1 mark for each word in the correct position For a database to be in First Normal Form (1NF) there must be no repeating groups of attributes. For a database to be in Second Normal Form (2NF), it must be in 1NF, and contain no partial key dependencies. For a database to be in Third Normal Form (3NF), it must be in 2NF, and all attributes must be fully dependent on the primary key.	4
----------	--	----------

QUESTION

- (d) Give **three** reasons why the EMPLOYEES database is fully normalised.

- 1
- 2
- 3

[3]

MARK SCHEME

3(d)	1 mark per bullet point to max 3 There are no repeating groups (1NF) There are no partial dependencies (2NF) There are no non-key dependencies // There are no transitive dependencies (3NF)	3
------	---	----------

9608_s18_qp_12_sc_7.c

QUESTION

(c) The database has been normalised to Third Normal Form (3NF).
 Define the three stages of database normalisation.

1NF

.....

2NF

.....

3NF

.....

[3]

MARK SCHEME

7(c)	Max 1 mark from each bulleted group 1NF No repeated groups of attributes All attributes should be atomic No duplicate rows 2NF (in 1NF and) No partial dependencies 3NF (in 2NF and) No non-key dependencies No transitive dependencies	3
------	--	----------

QUESTION

- 9 A database has been designed to store data about salespersons and the products they have sold.

The following facts help to define the structure of the database:

- each salesperson works in a particular shop
- each salesperson has a unique first name
- each shop has one or more salespersons
- each product which is sold is manufactured by one company only
- each salesperson can sell any of the products
- the number of products that each salesperson has sold is recorded

The table *ShopSales* was the first attempt at designing the database.

FirstName	Shop	ProductName	NoOfProducts	Manufacturer
Nick	TX	television set	3	SKC
		refrigerator	2	WP
		digital camera	6	HKC
Sean	BH	hair dryer	1	WG
		electric shaver	8	BG
John	TX	television set	2	SKC
		mobile phone	8	ARC
		digital camera	4	HKC
		toaster	3	GK

- (a) State why the table is **not** in First Normal Form (1NF).

.....
[1]

MARK SCHEME

- 9 (a) Any **one** from:

- (ShopSales) table has repeated group (of attributes)
- each sales person has a number of products
- FirstName, Shop would need to be repeated for each record

[1]

QUESTION

(b) The database design is changed to:

SalesPerson (FirstName, Shop)

SalesProducts (FirstName, ProductName, NoOfProducts, Manufacturer)

Using the data given in the first attempt table (ShopSales), show how these data are now stored in the revised table designs.

Table: SalesPerson

FirstName	Shop

Table: SalesProducts

FirstName	ProductName	NoOfProducts	Manufacturer

[3]

MARK SCHEME

(b) One mark for SalesPerson table

table: SalesPerson

FirstName	Shop
Nick	TX
Sean	BH
John	TX

table: SalesProducts

FirstName	ProductName	NoOfProducts	Manufacturer
Nick	television set	3	SKC
Nick	refrigerator	2	WP
Nick	digital camera	6	HKC
Sean	hair dryer	1	WG
Sean	electric shaver	8	BG
John	television set	2	SKC
John	mobile phone	8	ARC
John	digital camera	4	HKC
John	toaster	3	GK

(1 mark for FirstName column + 1 mark for remainder of table)

[3]

QUESTION

(iii) Write the table definitions to give the database in 3NF.

.....
.....
.....
.....[2]

MARK SCHEME

(iii) SalesPerson (FirstName, Shop)
-SalesProducts (FirstName, ProductName, NoOfProducts) **OR**
SalesProducts(SalesID, FirstName, ProductName, NoOfProducts)

-Product (ProductName, Manufacturer)

1 mark for correct attributes in SalesProducts and Product tables and **1 mark** for correct identification of both primary keys **[2]**

QUESTION

- 9 A database has been designed to store data about salespersons and the products they have sold.

The following facts help to define the structure of the database:

- each salesperson works in a particular shop
- each salesperson has a unique first name
- each shop has one or more salespersons
- each product which is sold is manufactured by one company only
- each salesperson can sell any of the products
- the number of products that each salesperson has sold is recorded

The table *ShopSales* was the first attempt at designing the database.

FirstName	Shop	ProductName	NoOfProducts	Manufacturer
Nick	TX	television set	3	SKC
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		digital camera	6	HKC
Sean	BH	hair dryer	1	WG
		electric shaver	8	BG
John	TX	television set	2	SKC
		mobile phone	8	ARC
		digital camera	4	HKC
		toaster	3	GK

- (a) State why the table is **not** in First Normal Form (1NF).

.....
[1]

MARK SCHEME

- 9 (a) Any **one** from:

- (ShopSales) table has repeated group (of attributes)
- each sales person has a number of products
- FirstName, Shop would need to be repeated for each record

[1]

QUESTION

(b) The database design is changed to:

SalesPerson (FirstName, Shop)

SalesProducts (FirstName, ProductName, NoOfProducts, Manufacturer)

Using the data given in the first attempt table (ShopSales), show how these data are now stored in the revised table designs.

Table: SalesPerson

FirstName	Shop

Table: SalesProducts

FirstName	ProductName	NoOfProducts	Manufacturer

[3]

MARK SCHEME

(b) One mark for SalesPerson table

table: SalesPerson

FirstName	Shop
Nick	TX
Sean	BH
John	TX

table: SalesProducts

FirstName	ProductName	NoOfProducts	Manufacturer
Nick	television set	3	SKC
Nick	refrigerator	2	WP
Nick	digital camera	6	HKC
Sean	hair dryer	1	WG
Sean	electric shaver	8	BG
John	television set	2	SKC
John	mobile phone	8	ARC
John	digital camera	4	HKC
John	toaster	3	GK

(1 mark for FirstName column + 1 mark for remainder of table)

[3]

QUESTION

(iii) Write the table definitions to give the database in 3NF.

.....
.....
.....
.....[2]

MARK SCHEME

(iii) SalesPerson (FirstName, Shop)
-SalesProducts (FirstName, ProductName, NoOfProducts) **OR**
SalesProducts(SalesID, FirstName, ProductName, NoOfProducts)

-Product (ProductName, Manufacturer)

1 mark for correct attributes in SalesProducts and Product tables and **1 mark** for correct identification of both primary keys **[2]**

Explain why a given set of database tables are, or are not, in 3NF

9608_s20_qp_12_sc_6.a

QUESTION

- 6 A software development company has a relational database, SOFTWARE_MANAGEMENT. The database stores details of the customers who have purchased software, as well as the software and licences that customers have purchased.

The SOFTWARE_MANAGEMENT database has the following tables:

CUSTOMER_DETAILS (CustomerID, CompanyName, Address1, Address2, City)

SOFTWARE_PURCHASED (SoftwareName, SoftwareDescription, CustomerID, LicenceType, LicenceCost, RenewalDate)

- (a) Explain why this database is **not** in Third Normal Form (3NF). Refer to the tables in your answer.

Do **not** attempt to normalise the tables.

.....

.....

.....

.....

.....

..... [2]

MARK SCHEME

6(a)	1 mark per bullet point to max 2 - There are partial dependencies in the SOFTWARE_PURCHASED table // SoftwareDescription is dependent only on SoftwareName and not both SoftwareName and CustomerID - There is a non-key dependency in the SOFTWARE_PURCHASED table // LicenceCost is dependent on LicenceType	2
------	--	----------

QUESTION

(d) Give **three** reasons why the `EMPLOYEES` database is fully normalised.

- 1
-
- 2
-
- 3
-

[3]

MARK SCHEME

3(d)	1 mark per bullet point to max 3 There are no repeating groups (1NF) There are no partial dependencies (2NF) There are no non-key dependencies // There are no transitive dependencies (3NF)	3
------	---	----------

QUESTION

(c) The database designer states that the `PURPLEGAME` database is in Third Normal Form (3NF).

Tick (✓) **one** box to indicate whether this statement is true or false.

True	False
☐	☐

Justify your choice.

-
-
-
-
-
-
-[3]

MARK SCHEME

2(c)	1 mark for the correct box ticked <table><tr><td>True</td><td>False</td></tr><tr><td>✓</td><td></td></tr></table> 1 mark per bullet for justification, max 2 No repeated attributes // data is atomic // No partial dependencies (no dual keys) No non-key / transitive dependencies	True	False	✓		3
True	False					
✓						

9608_s15_qp_12_sc_9.c.ii

QUESTION

(ii) Explain why the `SalesProducts` table is **not** in Third Normal Form (3NF).

.....

[2]

MARK SCHEME

- (ii) • There is a non-key dependency
 • Manufacturer is dependent on ProductName, (which is not the primary key of the SalesProducts table) **[2]**

9608_s15_qp_11_sc_9.c.ii

QUESTION

(ii) Explain why the `SalesProducts` table is **not** in Third Normal Form (3NF).

.....

[2]

MARK SCHEME

- (ii) • There is a non-key dependency
• Manufacturer is dependent on ProductName, (which is not the primary key of the SalesProducts table)

[2]

Produce a normalised database design for a description of a database, a given set of data, or a given set of tables

9608_w19_qp_13_sc_3.a.i

QUESTION

3 A hotel needs to record information about customers and their bookings.

(a) The hotel has two types of room: double and family. Each room has a unique room number.

The hotel stores information about the customers including their name, address and contact details.

When a customer books a room, they give the start date and the number of nights they want to stay. If a customer wants more than one room, each room must have a separate booking. Each booking has an ID number.

The hotel creates a normalised, relational database to store the required information. There are three tables:

- CUSTOMER
- ROOM
- BOOKING

(i) Complete the database design for the hotel by writing the attributes for each table.

CUSTOMER (.....
.....
.....)
ROOM (.....
.....
.....)
BOOKING (.....
.....
.....)

[3]

MARK SCHEME

3(a)(i)	1 mark per table CUSTOMER table has at least customer ID, customer name, address and contact details ROOM has at least room number, room type, BOOKING has at least booking ID, room number, customer ID, start date, number of nights CUSTOMER (<u>CustomerID</u>, Name, Address, ContactDetails) ROOM (RoomNumber, RoomType) BOOKING (BookingID, RoomNumber, CustomerID, StartDate, NumberNights)	3
---------	---	----------

9608_w19_qp_13_sc_3.a.ii

QUESTION

(ii) Identify the primary key for each table that you designed in **part (a)(i)**.

CUSTOMER

ROOM

BOOKING [2]

MARK SCHEME

3(a)(ii)	1 mark for 1 or 2 correct Primary Keys, 2 marks for 3 correct Primary Keys CUSTOMER: CustomerID ROOM: RoomNumber BOOKING: BookingID	2
----------	---	----------

9608_w19_qp_13_sc_3.a.iii

QUESTION

(iii) Identify **one** foreign key in the tables that you designed in **part (a)(i)**.

Table name

Foreign key [1]

MARK SCHEME

3(a)(iii)	1 mark for both table name and Foreign Key Table: <code>BOOKING</code> Foreign Key: <code>CustomerID / RoomNumber</code>	1
-----------	---	----------

9608_w19_qp_11_sc_4.d.i

QUESTION

- (d) The bank uses a relational database, `ACCOUNTS`, to store the information about customers and their accounts.

The database stores the customer's first name, last name and date of birth.

The bank has several different types of account. Each account type has a unique ID number, name (for example, regular or saving) and bonus (for example, \$5.00, \$10.00 or \$15.00).

A customer can have more than one account.

Each customer's account has its own ID number and stores the amount of money the customer has in that account.

The bank creates a normalised, relational database to store the required information. There are three tables:

- `CUSTOMER`
- `ACCOUNT_TYPE`
- `CUSTOMER_ACCOUNT`

- (i) Write the attributes for each table to complete the database design for the bank.

```

CUSTOMER ( .....
.....
.....)

ACCOUNT_TYPE ( .....
.....
.....)

CUSTOMER_ACCOUNT ( .....
.....
.....)
[3]

```

MARK SCHEME

4(d)(i)	1 mark per table Table CUSTOMER with fields FirstName, LastName, DateOfBirth, CustomerID Table ACCOUNT_TYPE with fields AccountID, Name, Bonus Table CUSTOMER_ACCOUNT with fields ID, CustomerID, AccountID, Amount CUSTOMER (<u>CustomerID</u> , FirstName, LastName, DateOfBirth) ACCOUNT_TYPE (<u>AccountID</u> , Name, Bonus) CUSTOMER_ACCOUNT (<u>ID</u> , CustomerID, AccountID, Amount)	3
---------	--	----------

9608_w19_qp_11_sc_4.d.ii

QUESTION

(ii) Identify the primary key for each table that you designed in **part (d)(i)**.

CUSTOMER

ACCOUNT_TYPE

CUSTOMER_ACCOUNT [2]

MARK SCHEME

4(d)(ii)	1 mark for 1 or 2 correct Primary Keys, 2 marks for 3 correct Primary Keys CUSTOMER: CustomerID ACCOUNT_TYPE: AccountID CUSTOMER_ACCOUNT: ID	2
----------	---	----------

9608_w19_qp_11_sc_4.d.iii

QUESTION

(iii) Identify **one** foreign key in one of the tables that you designed in **part (d)(i)**.

Table name

Foreign key [1]

MARK SCHEME

4(d)(iii)	1 mark for both table name and Foreign Key Table: CUSTOMER_ACCOUNT Foreign Key: CustomerID / AccountID	1
-----------	---	----------

9608_w17_qp_13_sc_7.b

QUESTION

(b) The doctors are concerned that many patients make appointments but do not attend them.

Describe the changes to the table designs that could be made to store this information.

.....

.....[2]

MARK SCHEME

7(b)	Two marks from: Either: Add an attribute (for example <i>Attended</i>) To the appointment table // <i>APPOINTMENT</i> Or: Add an attribute (for example <i>AppointmentsMissed</i>) To the patient table // <i>PATIENT</i>	2
------	--	----------

9608_w17_qp_13_sc_7.c.ii

QUESTION

(ii) Opening a new clinic in the neighbouring village will not require any additional table for storing appointments. It will need a change to the existing appointment table design.

Show the revised *APPOINTMENT* table.

APPOINTMENT (.....)

.....) [1]

MARK SCHEME

7(c)(ii)	<i>APPOINTMENT</i> (Site, AppointmentDate, AppointmentTime, DoctorID, PatientID)	1
----------	---	----------

9608_w17_qp_12_sc_7.b

QUESTION

- (b) The company decides to produce a visit report, `VisitReportText`, for each visit made.

This text will be produced from the one or more interview texts obtained at the visit.

State how one or more of the given table designs can be changed to add this attribute.

.....
.....[1]

MARK SCHEME

7(b)	1 Mark for correct answer Add attribute <code>VisitReportText</code> to table <u>VISIT</u>	1
------	--	----------

9608_w17_qp_12_sc_7.e

QUESTION

- (e) At present, all interviews are performed in the UK. Many clients now operate in other countries in Europe. The company wants to perform interviews with the client's staff in other countries. Not all interview staff are willing to travel outside of the UK.

State how one or more of the table designs should be revised to store this information.

.....
.....[1]

MARK SCHEME

7(e)	1 Mark for a correct answer Add a suitable attribute, for example, <code>EuropeTraveller</code> to the <u>STAFF</u> table // Add a suitable attribute, for example, <code>Country</code> to the <u>CLIENT</u> table	1
------	---	----------

9608_s17_qp_13_sc_1.b.i

QUESTION

- (b) A relational database is created to store the ward and nurse data. The two table designs for Area A are:

A-WARD (WardName, NumberOfBeds)

A-NURSE (NurseID, FirstName, FamilyName,)

- (i) Complete the design for the A-NURSE table. [1]

MARK SCHEME

1(b)(i)	A-NURSE (<u>NurseID</u> , FirstName, FamilyName, WardName)	1
---------	---	---

9608_s17_qp_13_sc_1.c.ii

QUESTION

(ii) The design for the Area B data is as follows:

B-NURSE (NurseID, FirstName, FamilyName, Specialism)

B-WARD (WardName, NumberOfBeds, Specialism)

B-WARD-NURSE (.....)

Complete the attributes for the third table. Underline its primary key. [2]

MARK SCHEME

1(c)(ii)	B-WARD-NURSE (<u>WardName</u> , <u>NurseID</u>)	2
	Both attributes (with no additions)	1
	Joint primary key correctly underlined	1

9608_s17_qp_11_sc_1.b.i

QUESTION

(b) A relational database is created to store the ward and nurse data. The two table designs for Area A are:

A-WARD (WardName, NumberOfBeds)

A-NURSE (NurseID, FirstName, FamilyName,)

(i) Complete the design for the A-NURSE table. [1]

MARK SCHEME

1(b)(i)	A-NURSE (<u>NurseID</u> , FirstName, FamilyName, WardName)	1
---------	---	---

QUESTION

(ii) The design for the Area B data is as follows:

B-NURSE (NurseID, FirstName, FamilyName, Specialism)

B-WARD (WardName, NumberOfBeds, Specialism)

B-WARD-NURSE (.....)

Complete the attributes for the third table. Underline its primary key. [2]

MARK SCHEME

1(c)(ii)	B-WARD-NURSE (<u>WardName</u> , <u>NurseID</u>)	2
	Both attributes (with no additions)	1
	Joint primary key correctly underlined	1

QUESTION

(b) The database design is changed to:

SalesPerson (FirstName, Shop)

SalesProducts (FirstName, ProductName, NoOfProducts, Manufacturer)

Using the data given in the first attempt table (ShopSales), show how these data are now stored in the revised table designs.

Table: SalesPerson

FirstName	Shop

Table: SalesProducts

FirstName	ProductName	NoOfProducts	Manufacturer

[3]

MARK SCHEME

(b) One mark for SalesPerson table

table: SalesPerson

FirstName	Shop
Nick	TX
Sean	BH
John	TX

table: SalesProducts

FirstName	ProductName	NoOfProducts	Manufacturer
Nick	television set	3	SKC
Nick	refrigerator	2	WP
Nick	digital camera	6	HKC
Sean	hair dryer	1	WG
Sean	electric shaver	8	BG
John	television set	2	SKC
John	mobile phone	8	ARC
John	digital camera	4	HKC
John	toaster	3	GK

(1 mark for FirstName column + 1 mark for remainder of table)

[3]

9608_s15_qp_12_sc_9.c.iii

QUESTION

(iii) Write the table definitions to give the database in 3NF.

.....
.....
.....
.....[2]

MARK SCHEME

(iii) SalesPerson (FirstName, Shop)
-SalesProducts (FirstName, ProductName, NoOfProducts) **OR**
SalesProducts(SalesID, FirstName, ProductName, NoOfProducts)

-Product (ProductName, Manufacturer)

1 mark for correct attributes in SalesProducts and Product tables and **1 mark** for correct identification of both primary keys **[2]**

9608_s15_qp_11_sc_9.c.iii

QUESTION

(iii) Write the table definitions to give the database in 3NF.

.....
.....
.....
.....[2]

MARK SCHEME

(iii) SalesPerson (FirstName, Shop)
-SalesProducts (FirstName, ProductName, NoOfProducts) **OR**
SalesProducts(SalesID, FirstName, ProductName, NoOfProducts)

-Product (ProductName, Manufacturer)

1 mark for correct attributes in SalesProducts and Product tables and **1 mark** for correct identification of both primary keys **[2]**